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PART I

Amendment 8: Where the Leaked Signal Lives

1

Why a follow-up, and what was known

Study 2 established *that* crop-level splitting inflates accuracy and that the inflation is leakage, but neither registered mechanism test isolated *how* models use it. Two observations pointed the way. Leak models kept most of their advantage on backgrounds transplanted from unseen fields, which suggested a carrier inside the cell region rather than in the background. And the ConvNeXt-T adversary had been evaded, so the gradient-reversal arm had only one usable architecture.

Amendment 8 (`PREREGISTRATION_STUDY2_AMENDMENT8.md`, commit `80171fe`, 2026-10-19 14:02 IST) registered eight new hypotheses after the Study 2 analysis and before any follow-up model was trained or any follow-up analysis was run on existing outputs. It states what was known and what was not: every registered Study 2 output was known, including transplant condition means and gradient-reversal sweep means; no per-cell CRIC result by slide composition, no ablated image evaluation, no erased-feature evaluation, no Grad-CAM map, and no model trained with masked backgrounds, colour augmentation, stain normalisation, or the linear adversary had been seen. F1–F8 form their own Holm family, separate from H1–H5, whose verdicts the amendment does not touch.

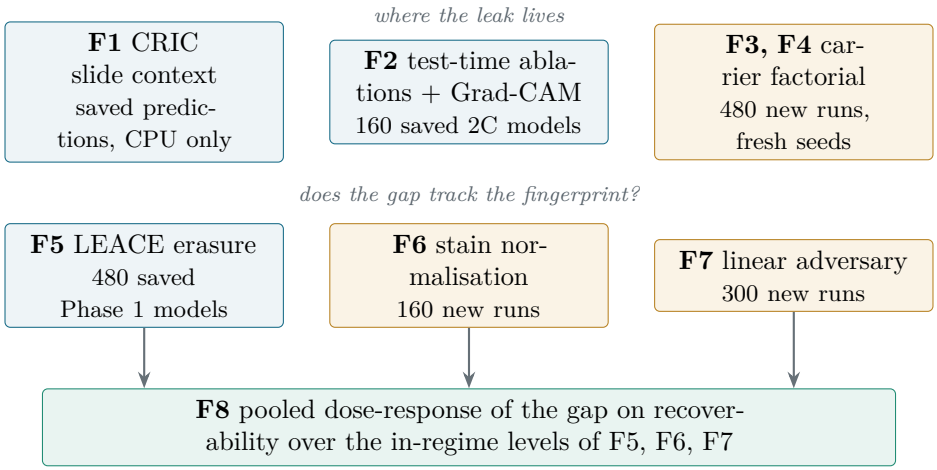


Figure 1.1: The eight Amendment 8 hypotheses. Blue arms reuse saved Study 2 artifacts (no retraining); gold arms train new models on seeds no earlier arm used; F8 pools the three dose-response arms.

2

The eight follow-up designs

2.1 F1: slide context in CRIC

CRIC slide images contain cells of several classes (83% of slides have more than one), so a test cell's training siblings can carry a different label. For every cell, the *context class* is the majority class of the other cells of its slide (cells whose slide has no other cell, or a tied majority, are excluded). A *discordant* cell's class differs from its context class. If crop-level models apply a slide-level label prior, discordant cells should receive more probability on the context class when their slide was in training (P1) than when it was not (P2). F1 is that within-cell difference over the 120 saved CRIC P1 and P2 runs, with a percentile cluster bootstrap over slides (10,000 replicates).

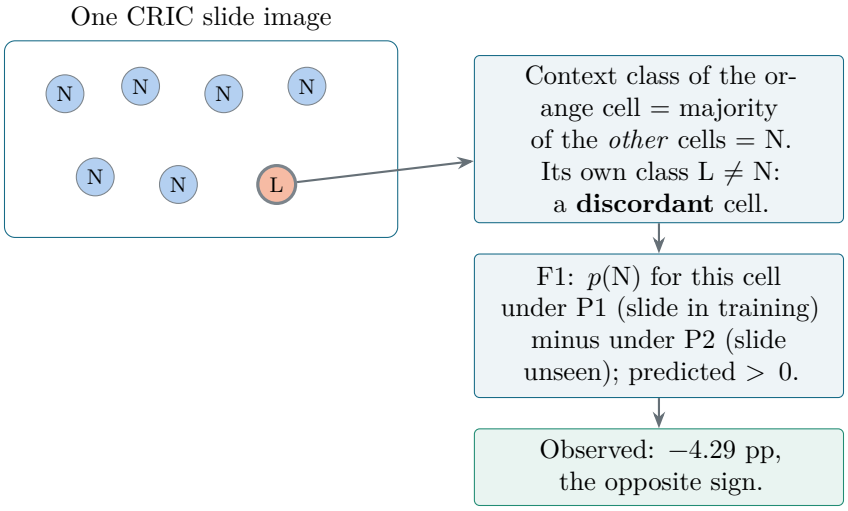


Figure 2.1: How F1 defines a discordant cell. A slide-level label prior would pull discordant cells toward their slide's majority class when the slide was seen in training; the models did the opposite.

2.2 F2: carrier ablations at test time

Each of the 160 saved 2C models is re-evaluated on its seed's common test set under six conditions rendered at native resolution: original; greyscale; Reinhard stain transfer of the crop's LAB mean and standard deviation to those of the 2B unseen same-class donor [5]; low-pass (Gaussian $\sigma = 2$ px at the 224-px input); illumination flattening (subtracting a $\sigma = 16$ px Gaussian); and background replaced by the ImageNet mean. F2 asks whether stain transfer shrinks the leak-minus-clean

advantage. Grad-CAM [7] attribution outside the cell mask is measured on the convolutional models and reported descriptively, with a random-head sanity check [1].

2.3 F3 and F4: carrier removal in training

The 2C factorial is rerun on ten seeds never used for any model (1103, 1151, 1201, 1249, 1301, 1361, 1409, 1451, 1499, 1553) under three conditions: the standard recipe; every training and test crop's background masked to the ImageNet mean; and strong HED optical-density colour augmentation [6, 8] ($\alpha \in [0.8, 1.2]$, $\beta \in [-0.2, 0.2]$ per stain channel, then brightness and contrast in $[0.8, 1.2]$) during training. F3 asks whether leakage survives with no background; F4 whether colour augmentation shrinks it. The standard condition is an independent replication of H3.

2.4 F5: rank-limited LEACE erasure

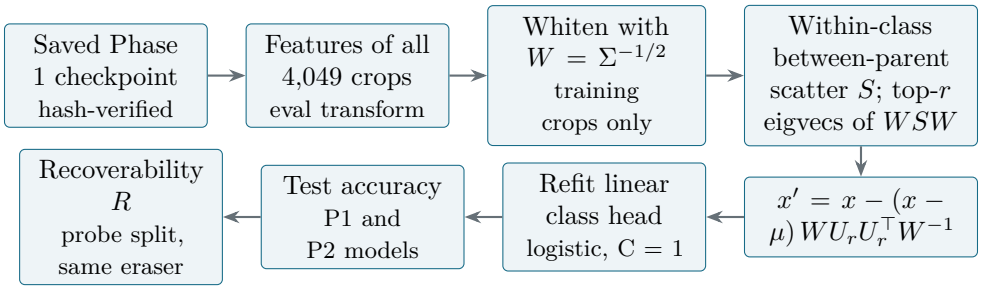


Figure 2.2: Rank-limited LEACE erasure (F5). At full rank the projection is LEACE for the within-class parent indicator: no linear function of the erased features covaries with it on the fitting data. Ranks 0, 4, 16, 64, and 256 give the dose.

Every saved Phase 1 P1 and P2 model (480) is re-evaluated: features of all crops are extracted, the within-class parent directions of the model's own training crops are erased at ranks 0, 4, 16, 64, and 256 [2], the class head is refitted, and test accuracy and recoverability are measured. The regime and manipulation check follow H5. F5 asks whether erasure at the strongest in-regime rank reduces the P1 − P2 gap.

2.5 F6: stain-normalisation dose

Every crop is Reinhard-normalised toward one dataset-wide reference at strength 0.5 and 1.0, in training, test, and probe crops alike. ResNet-50, ConvNeXt-T, DenseNet-121, and EfficientNet-B0 are trained on P1 and P2 for the ten Study 2 seeds no mechanism arm used; strength 0 is the matching Phase 1 run. F6 is the reduction in gap per unit of strength.

2.6 F7: probe-matched linear adversary

The 2A design with a different adversary (Figure 24.4): per-feature standardisation and one linear layer, its own optimiser at 10^{-3}, re-initialised every five epochs. ResNet-50, ConvNeXt-T, and DenseNet-121; $\lambda \in \{0, 0.1, 0.3, 1, 3\}$; P1 and P2; the same ten seeds as F6. F7 is the slope of gap on recoverability, as for H5.

2.7 F8: pooled dose-response

All included architecture-by-level cells of F5, F6, and F7, after their regimes and manipulation checks: gap $\sim$ cell + R_{pair} + (1 | manipulation:seed). F8 is the slope.

2.8 Recoverability, measured the same way everywhere

R for F5–F8 is computed with one GPU implementation of the registered logistic probe (sklearn's objective, C = 1, standardised features, L-BFGS). On the 280 Study 2 gradient-reversal runs it agreed with the registered sklearn probe at $r = 0.999$ (Figure 2.3).

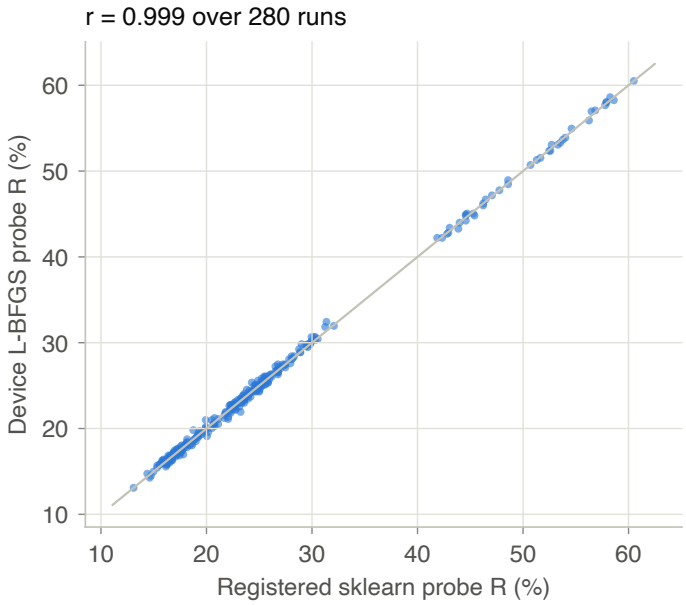


Figure 2.3: Within-class recoverability from the registered sklearn probe and the device probe used for F5–F8, on the 280 Study 2 adversarial runs.

3

Execution and the Amendment 9 fix

The new code was unit-tested on the workstation and, for GPU paths, on the host (13 tests, including a check that full-rank erasure leaves no covariance between features and within-class parent identity, and that the device probe matches sklearn to 10^{-3}). A synthetic dry run of the registered models recovered every planted effect. Eleven smoke tasks ran only on leak cells and single P1 runs; they confirmed that re-evaluated checkpoints reproduce their recorded accuracy and that the linear adversary learns. The caches for masking, normalisation, and the ablations were built and hashed, and the registration commit recorded every hash. The matrix of 1,580 tasks then ran from 19 Oct 14:03 to 20 Oct 02:52 without a failure.

Amendment 9: the preparer bug

When the registered models were first run on the completed carrier arm, the R script stopped at F4: the preparer had labelled all 160 standard-recipe runs as colour-augmented, because a missing colour augmentation (`None` in the run record) became a NaN in a DataFrame row, and NaN is truthy. The script halted before writing any output, so no F3 or F4 estimate was seen. The condition is now read from each run record, a unit test covers it, and the fix was registered as Amendment 9 (20 October, 01:25) before the corrected table was analysed. The synthetic dry run had bypassed the preparer, which is why the bug was not caught earlier.

4

Follow-up results

Table 4.1: The eight registered follow-up tests (one Holm family). Estimates are oriented so that the predicted direction is positive.

	Question	Estimate	95% CI	Holm p	Verdict
F1	CRIC pull toward the slide's majority class	−4.29	−8.60 to −1.37	0.001	**not supported**
F2	Stain transfer shrinks the leak effect	1.27	0.38 to 2.16	0.03	**supported**
F3	Leak effect with backgrounds masked	1.86	1.42 to 2.30	<0.001	**supported**
F4	Colour augmentation shrinks the leak effect	−0.23	−0.81 to 0.35	1.00	**not supported**
F5	LEACE erasure shrinks the gap	−0.01	−0.10 to 0.07	1.00	**not supported**
F6	Stain normalisation shrinks the gap	0.02	−0.41 to 0.44	1.00	**not supported**
F7	Gap tracks R (linear adversary)	0.000	−0.038 to 0.039	1.00	**not supported**
F8	Gap tracks R (pooled)	0.000	−0.005 to 0.005	1.00	**not supported**

4.1 The leak needs no background (F3)

Trained and tested with every background replaced by a constant, leak models beat clean models by 1.86 pp (1.42–2.30; Holm p <0.001), at least as much as under the standard recipe on the same fresh seeds (1.47 pp), which also replicates H3 (1.47 pp, 0.06–2.89). Masking changed the leak effect by −0.39 pp relative to standard training (−0.96 to 0.19). At test time, removing the background from the original 2C models' inputs left 68% of their advantage.

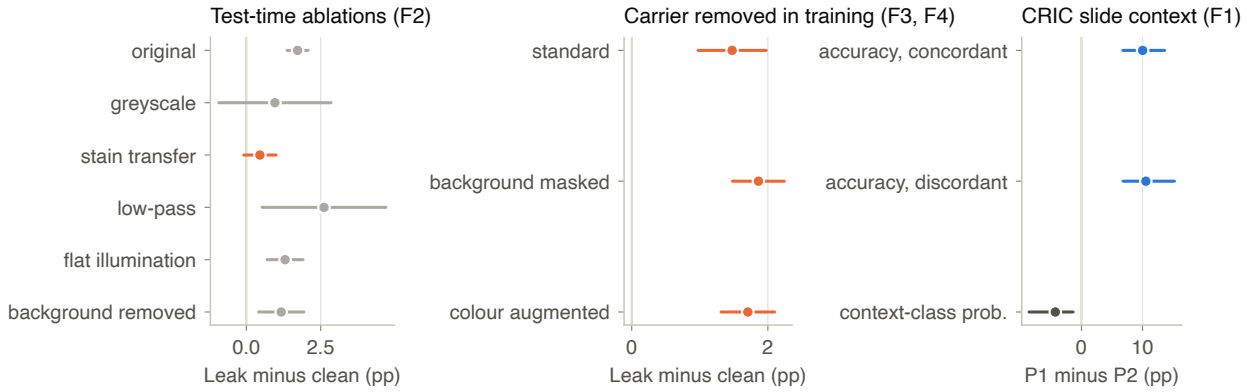


Figure 4.1: Left: the leak advantage under each test-time ablation (F2 highlighted). Middle: the leak effect under each training condition on fresh seeds (F3, F4). Right: CRIC within-cell accuracy gains by concordance, and the context-class probability for discordant cells (F1).

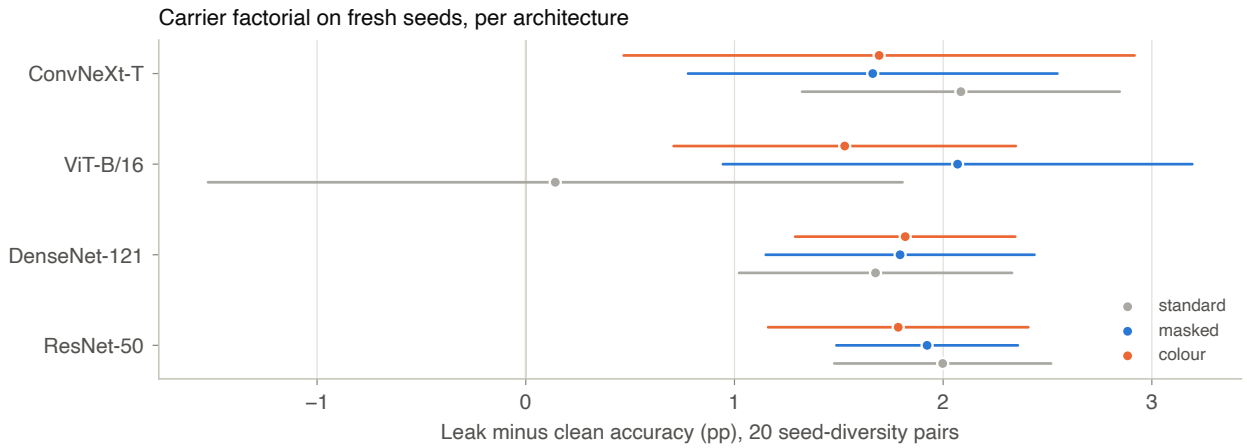


Figure 4.2: The carrier factorial per architecture.

Table 4.2: Amendment 8 carrier factorial (fresh seeds, 10 per cell): accuracy per cell (percent) and the leak effect (pp).

Architecture	Condition	Leak high	Clean high	Leak low	Clean low	Leak effect
ResNet-50	standard	96.06 ± 1.42	93.99 ± 2.08	96.40 ± 0.79	94.48 ± 1.52	2.00
	masked	95.46 ± 1.19	93.42 ± 1.26	95.69 ± 1.36	93.87 ± 1.37	1.92
	colour	95.62 ± 1.29	93.79 ± 1.29	95.84 ± 1.04	94.10 ± 1.45	1.78
DenseNet-121	standard	96.33 ± 1.16	95.14 ± 1.63	97.01 ± 0.66	94.85 ± 1.50	1.68
	masked	95.59 ± 1.32	94.21 ± 1.00	96.19 ± 0.87	93.99 ± 1.84	1.79
	colour	96.20 ± 1.01	94.47 ± 1.48	96.45 ± 0.58	94.54 ± 1.68	1.82
ViT-B/16	standard	93.40 ± 3.70	93.60 ± 1.77	94.97 ± 1.74	94.49 ± 1.32	0.14
	masked	94.92 ± 2.21	92.95 ± 1.97	94.91 ± 1.13	92.74 ± 1.56	2.07
	colour	95.05 ± 1.68	93.53 ± 1.27	95.04 ± 1.67	93.51 ± 1.20	1.53
ConvNeXt-T	standard	96.56 ± 0.95	95.05 ± 1.23	96.66 ± 0.67	94.00 ± 2.32	2.08
	masked	94.80 ± 1.73	93.15 ± 2.00	95.29 ± 1.47	93.61 ± 1.74	1.66
	colour	95.30 ± 1.77	94.24 ± 1.95	95.77 ± 1.51	93.44 ± 2.68	1.69

4.2 The leak is tied to the field's colour rendering at test time (F2), but colour augmentation does not prevent it (F4)

Moving each test cell's colour statistics to those of an unseen field of the same class cut the leak advantage from 1.72 to 0.46 pp, a reduction of 1.27 pp (0.38–2.16; Holm $p = 0.03$), removing 74% of it. Interventions

with comparable accuracy costs did not: with illumination flattened 76% of the advantage remained, and with the background removed 68%. Greyscale and low-pass filtering degraded accuracy too far to be informative (Figure 4.3). Yet models trained with strong colour augmentation leaked as much as standard models (1.71 versus 1.47 pp; F4 −0.23, −0.81 to 0.35).

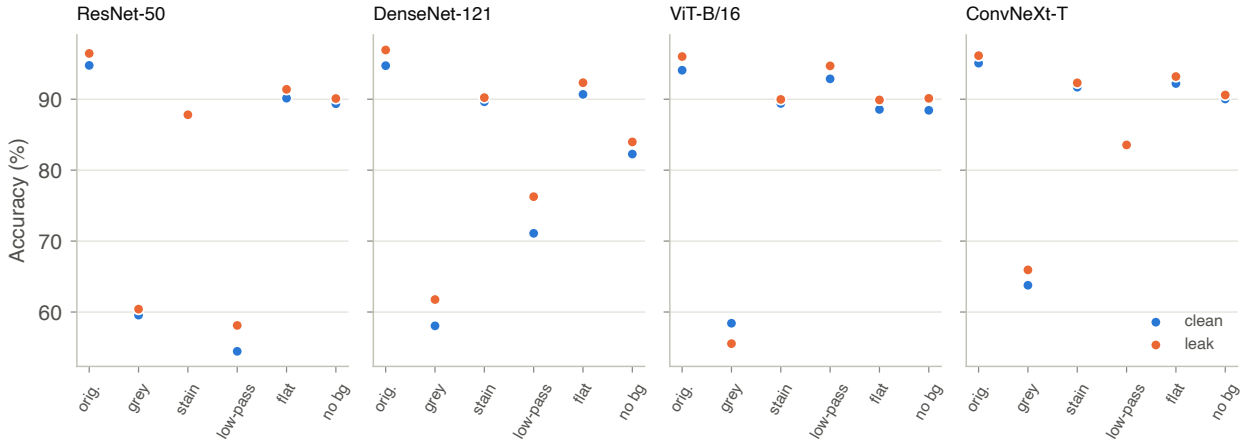


Figure 4.3: Accuracy of leak and clean 2C models under each test-time ablation, per architecture.

Table 4.3: Amendment 8 test-time carrier ablations on the Study 2 2C models: leak and clean accuracy (percent, 20 runs each) and the leak advantage (pp).

Architecture	Ablation	Leak models	Clean models	Leak − clean
ResNet-50	original	96.45 ± 0.94	94.77 ± 1.33	1.69
	grey	60.41 ± 5.61	59.56 ± 7.27	0.84
	stain_transfer	87.81 ± 2.85	87.76 ± 2.77	0.05
	low_pass	58.11 ± 11.63	54.46 ± 11.17	3.65
	illumination_flat	91.39 ± 2.88	90.15 ± 2.18	1.25
	background_grey	90.10 ± 2.57	89.36 ± 1.84	0.74
DenseNet-121	original	96.94 ± 0.78	94.73 ± 1.50	2.21
	grey	61.76 ± 6.30	58.05 ± 8.26	3.71
	stain_transfer	90.23 ± 2.21	89.64 ± 1.92	0.59
	low_pass	76.26 ± 6.79	71.09 ± 5.97	5.17
	illumination_flat	92.33 ± 2.17	90.68 ± 3.18	1.64
	background_grey	83.98 ± 3.72	82.27 ± 5.20	1.71
ViT-B/16	original	96.01 ± 1.15	94.08 ± 2.01	1.93
	grey	55.54 ± 5.12	58.42 ± 7.25	−2.87
	stain_transfer	89.97 ± 2.40	89.40 ± 3.25	0.57
	low_pass	94.70 ± 1.74	92.87 ± 2.95	1.83
	illumination_flat	89.90 ± 2.63	88.56 ± 3.56	1.34
	background_grey	90.12 ± 2.73	88.44 ± 3.54	1.68
ConvNeXt-T	original	96.14 ± 1.38	95.08 ± 1.49	1.06
	grey	65.94 ± 5.38	63.77 ± 5.85	2.16
	stain_transfer	92.31 ± 2.68	91.69 ± 2.52	0.61
	low_pass	83.55 ± 7.47	83.76 ± 6.82	−0.20
	illumination_flat	93.20 ± 1.62	92.21 ± 1.87	0.98
	background_grey	90.59 ± 2.46	90.03 ± 1.98	0.56

4.3 Exploratory: not shared pixels

In leak cells a third or more of the test crops overlap a training sibling in the source image. Splitting the leak advantage by overlap (not registered) shows it is as large for crops that share no pixel with any training crop (1.68 pp) as for crops that do (1.78 pp), in the Study 2 models and in all three carrier conditions; stain transfer removed it in both subsets (0.70 and 0.08 pp). The leak generalises to new cells from a seen field rather than reflecting memorised pixels.

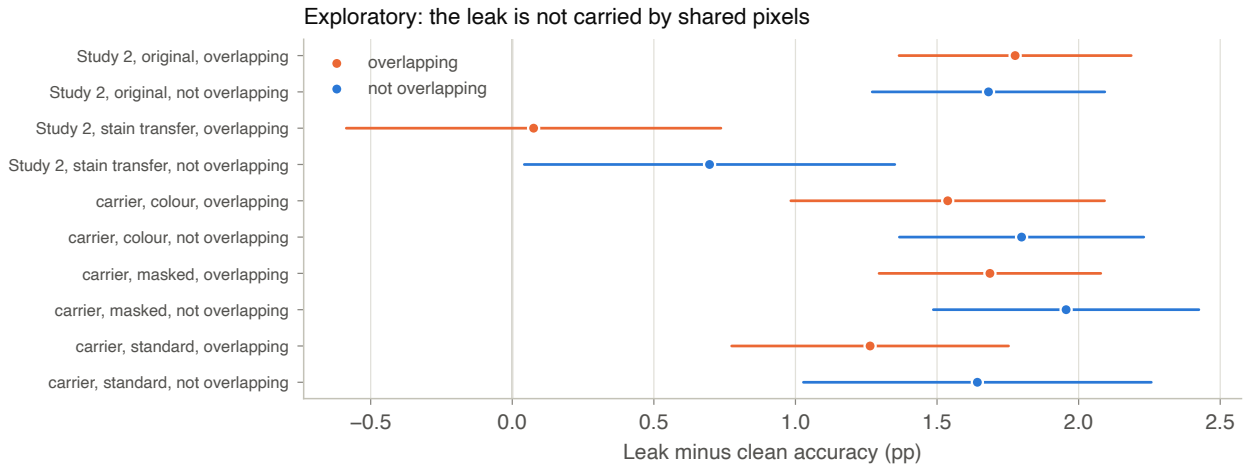


Figure 4.4: Exploratory: the leak advantage on test crops that do or do not share source pixels with a training sibling.

Table 4.4: EXPLORATORY (not registered): leak minus clean accuracy on test crops that do or do not share source pixels with a training sibling (pp; 95% t interval over architecture–seed–diversity pairs).

Condition	Subset	Pairs	Crops	Leak − clean	95% CI
Study 2, original	not overlapping	80	320	1.68	1.27 to 2.09
Study 2, original	overlapping	80	207	1.78	1.37 to 2.19
Study 2, stain transfer	not overlapping	80	320	0.70	0.04 to 1.35
Study 2, stain transfer	overlapping	80	207	0.08	−0.59 to 0.74
carrier, colour	not overlapping	80	329	1.80	1.37 to 2.23
carrier, colour	overlapping	80	206	1.54	0.98 to 2.09
carrier, masked	not overlapping	80	329	1.96	1.49 to 2.42
carrier, masked	overlapping	80	206	1.69	1.30 to 2.08
carrier, standard	not overlapping	80	329	1.64	1.03 to 2.26
carrier, standard	overlapping	80	206	1.26	0.77 to 1.75

4.4 No slide-level label prior (F1)

Over 1,144 discordant CRIC cells from 230 slides, the probability of the slide's majority class was −4.29 pp *lower*, not higher, when the cell's slide was in training (−8.60 to −1.37). All six architectures showed the same sign. Crop-level training raised accuracy as much for discordant cells (10.55 pp) as for concordant ones (10.02 pp; contrast −0.52 pp, −5.75 to 4.66): seen slides help each cell toward its own class, not toward the slide's typical class.

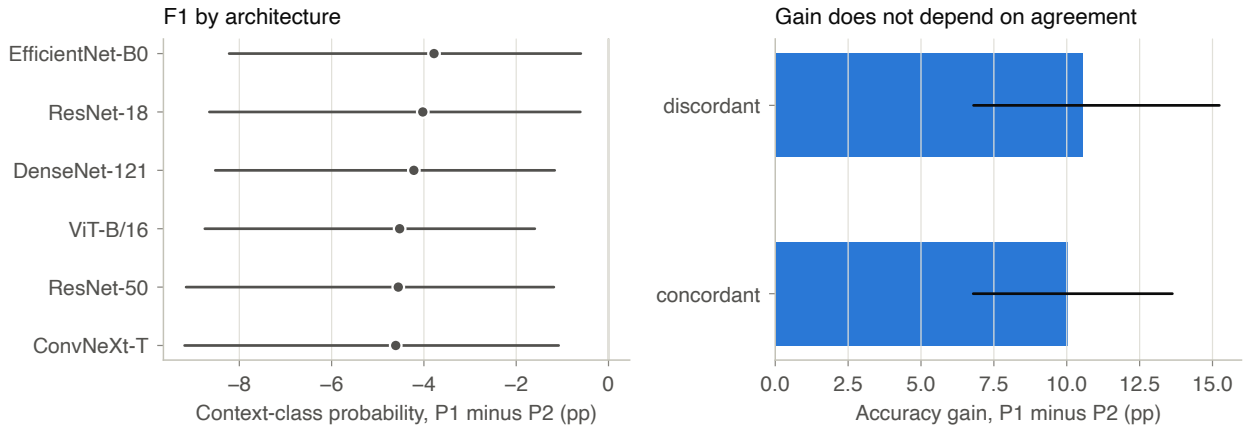


Figure 4.5: F1 per architecture, and the within-cell accuracy gain for concordant and discordant cells.

Table 4.5: F1 per architecture (secondary): within-cell P1 − P2 context-class probability on discordant cells (pp) with slide-bootstrap intervals.

Architecture	Estimate	95% CI	p	Cells
ConvNeXt-T	−4.61	−9.19 to −1.08	0.003	1,144
DenseNet-121	−4.22	−8.52 to −1.17	0.003	1,144
EfficientNet-B0	−3.78	−8.22 to −0.60	0.01	1,144
ResNet-18	−4.02	−8.65 to −0.61	0.01	1,144
ResNet-50	−4.56	−9.15 to −1.18	0.003	1,144
ViT-B/16	−4.53	−8.75 to −1.60	<0.001	1,144

4.5 The gap does not track the fingerprint (F5–F8)

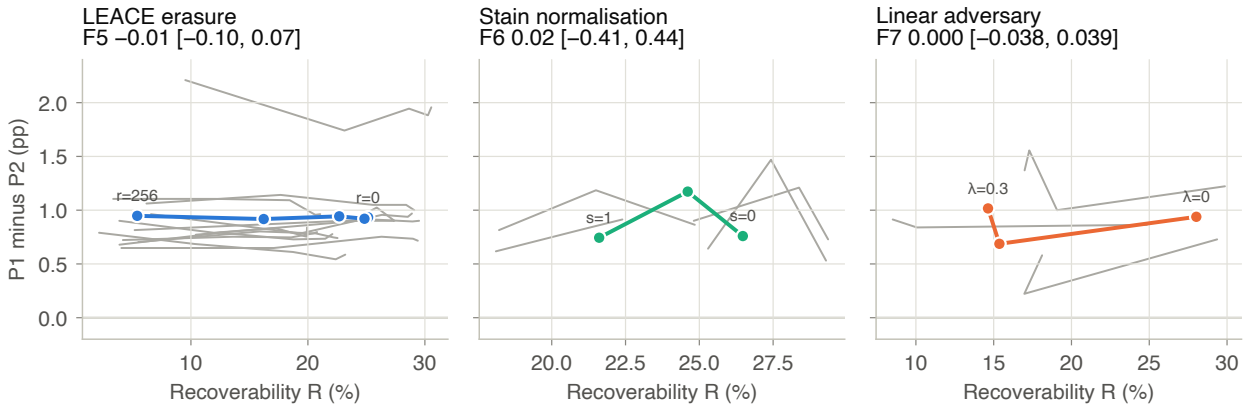


Figure 4.6: The P1 − P2 gap against recoverability along three manipulations (grey: architectures; colour: pooled over architectures that passed the manipulation check, at levels they share; registered in-regime levels only).

Rank-256 LEACE erasure cut recoverability from 25.1% to 5.4% (floor 1.2%) in all twelve architectures while costing P2 0.17 pp, and the gap stayed at 0.93 to 0.95 pp (F5 −0.01, −0.10 to 0.07). The probe-matched adversary passed its check in all three architectures, lowering R from 24.8–29.9% to 8.5–18.1% inside the regime, with a slope of 0.000 (−0.038 to 0.039). Stain normalisation lowered R only from 26.5% to 21.6% while costing P2 1.54 pp, failed its check for every architecture, and left the gap unchanged (F6 0.02, −0.41

to 0.44). Pooled over the 15 included cells, the slope was 0.000 (−0.005 to 0.005): even a 20-pp loss of recoverability would move the gap by at most about 0.1 pp.

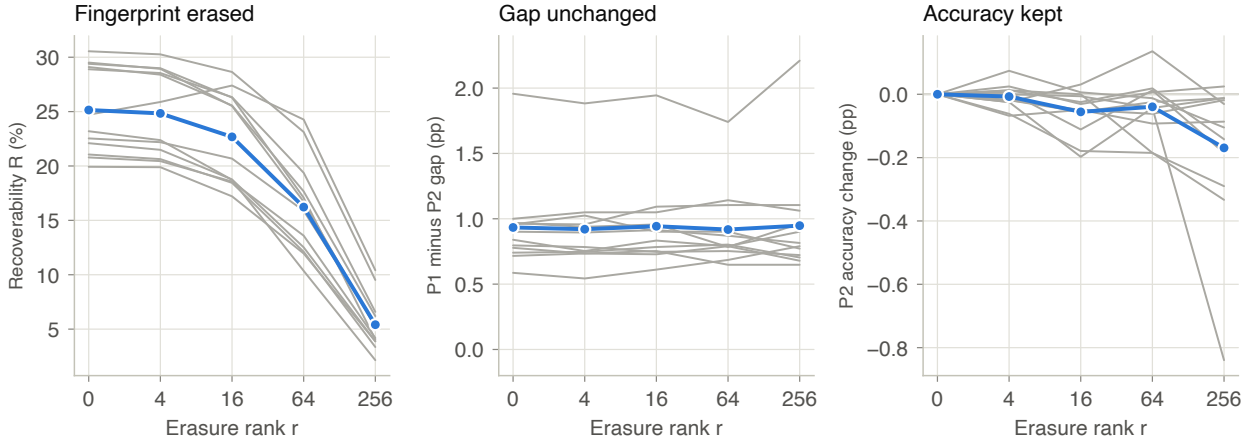


Figure 4.7: LEACE per architecture (grey) and pooled (blue): recoverability, gap, and P2 accuracy change by erasure rank.

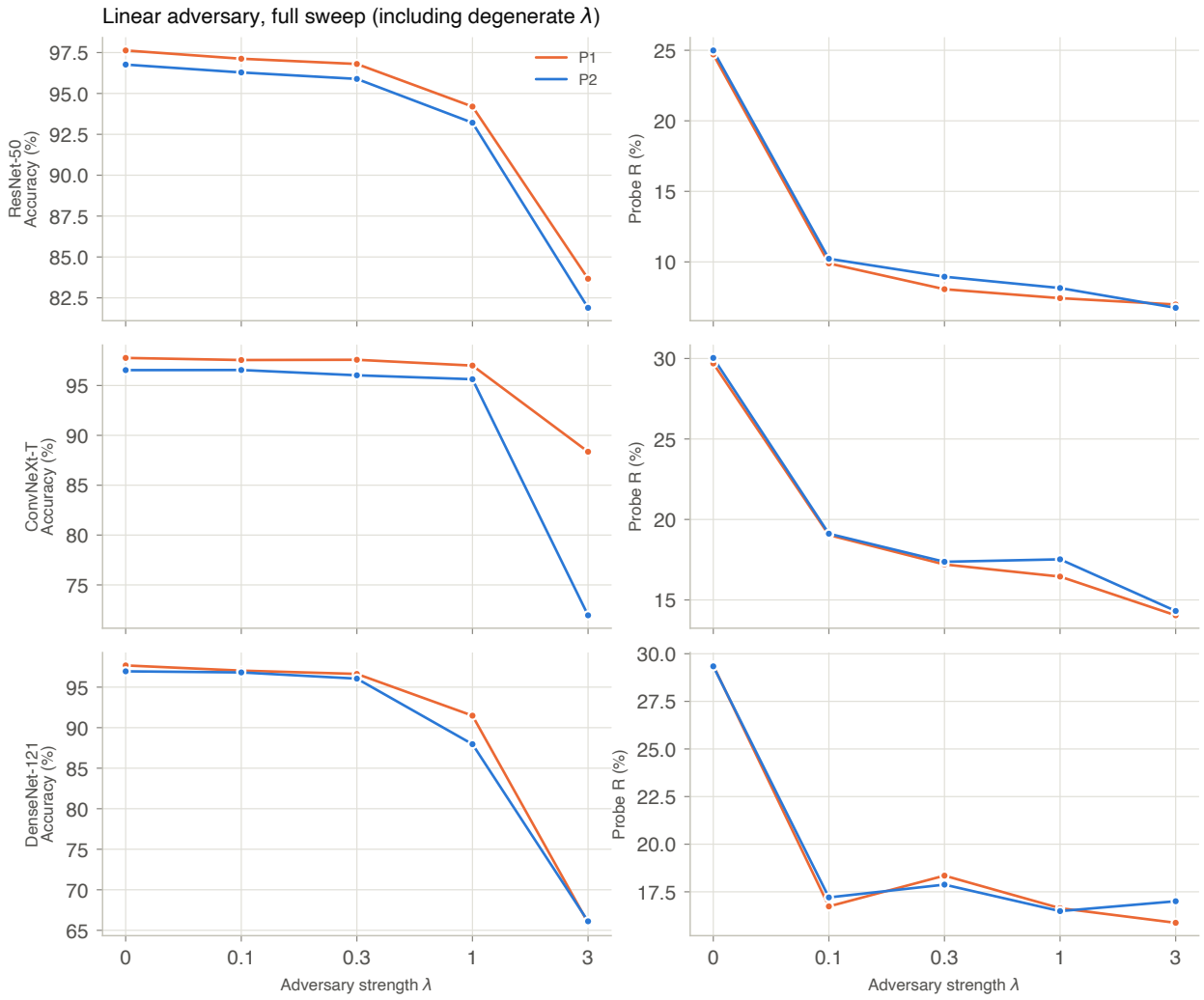


Figure 4.8: The full linear-adversary sweep per architecture, including the degenerate strengths ($\lambda = 1$ for ResNet-50 and DenseNet-121, $\lambda = 3$ for all) where P2 accuracy collapses and which the regime rule excludes.

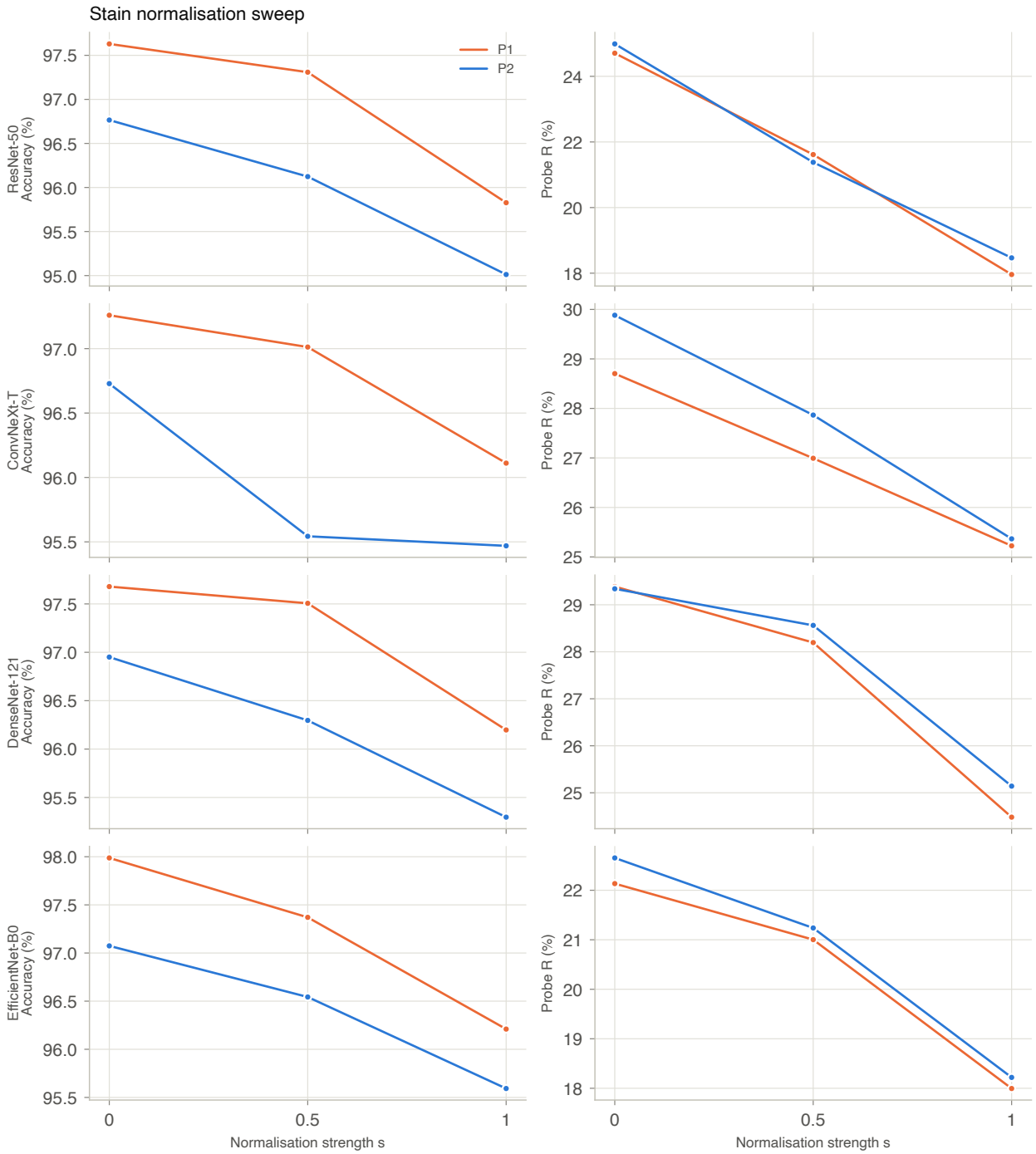


Figure 4.9: The stain-normalisation sweep per architecture: accuracy falls for both protocols, recoverability falls only modestly.

Table 4.6: Regimes and manipulation checks of every dose-response arm: levels whose mean P2 accuracy stays within 1 pp of baseline, the strongest such level, recoverability at baseline and there (percent), the class-only floor, and the check (at least a quarter of the above-floor R removed).

Arm	Architecture	Levels in regime	Strongest	R_0	R^*	Floor	Check
H5 GRL	ResNet-50	0, 0.01, 0.03, 0.1, 0.3, 1, 3	3	24.7	17.6	1.2	**PASS**
	ConvNeXt-T	0, 0.01, 0.03, 0.1, 0.3, 1, 3	3	29.5	55.1	1.2	**FAIL**
F5 LEACE	ResNet-18	0, 4, 16, 64, 256	256	23.2	2.1	1.2	**PASS**
	ResNet-50	0, 4, 16, 64, 256	256	24.7	10.4	1.2	**PASS**

Arm	Architecture	Levels in regime	Strongest	R_0	R^*	Floor	Check
	DenseNet-121	0, 4, 16, 64, 256	256	29.4	4.1	1.2	**PASS**
	EfficientNet-B0	0, 4, 16, 64, 256	256	22.5	5.5	1.2	**PASS**
	MobileNet-V2	0, 4, 16, 64, 256	256	20.8	4.2	1.2	**PASS**
	ViT-B/16	0, 4, 16, 64, 256	256	29.1	6.2	1.2	**PASS**
	ConvNeXt-T	0, 4, 16, 64, 256	256	29.5	5.2	1.2	**PASS**
	Swin-T	0, 4, 16, 64, 256	256	28.9	6.6	1.2	**PASS**
	RegNetY-1.6GF	0, 4, 16, 64, 256	256	21.1	3.4	1.2	**PASS**
	EfficientNetV2-S	0, 4, 16, 64, 256	256	19.9	3.9	1.2	**PASS**
	MaxViT-T	0, 4, 16, 64, 256	256	22.1	3.9	1.2	**PASS**
	DINOv2 ViT-S/14	0, 4, 16, 64, 256	256	30.5	9.5	1.2	**PASS**
F6 normalisation	ResNet-50	0, 0.5	0.5	24.8	21.5	1.2	**FAIL**
	DenseNet-121	0, 0.5	0.5	29.4	28.4	1.2	**FAIL**
	EfficientNet-B0	0, 0.5	0.5	22.4	21.1	1.2	**FAIL**
	ConvNeXt-T	0	0	29.3	29.3	1.2	**FAIL**
F7 linear adv.	ResNet-50	0, 0.1, 0.3	0.3	24.8	8.5	1.2	**PASS**
	DenseNet-121	0, 0.1, 0.3	0.3	29.4	18.1	1.2	**PASS**
	ConvNeXt-T	0, 0.1, 0.3, 1	1	29.9	17.0	1.2	**PASS**

Table 4.7: Amendment 8 rank-limited LEACE erasure on every Phase 1 model (20 seeds): accuracy with a refitted head (percent), gap (pp), logistic recoverability R, and nearest-centroid recoverability (percent).

Architecture	Rank	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)	NCC
ResNet-18	0	97.90 ± 0.37	97.31 ± 0.61	0.59	23.1 ± 1.2	23.2 ± 1.1	21.0 ± 1.0
	4	97.86 ± 0.37	97.32 ± 0.68	0.54	22.2 ± 1.2	22.5 ± 1.1	19.6 ± 1.1
	16	97.92 ± 0.38	97.31 ± 0.70	0.61	18.7 ± 1.1	18.7 ± 1.0	15.2 ± 1.1
	64	97.81 ± 0.37	97.12 ± 0.81	0.69	10.5 ± 0.8	10.2 ± 1.0	8.1 ± 0.8
	256	97.77 ± 0.33	96.98 ± 1.00	0.79	2.0 ± 0.4	2.3 ± 0.5	1.8 ± 0.4
ResNet-50	0	98.12 ± 0.42	97.16 ± 0.67	0.96	24.6 ± 1.1	24.9 ± 1.1	23.0 ± 1.2
	4	98.12 ± 0.45	97.09 ± 0.72	1.02	25.9 ± 1.5	25.9 ± 1.2	23.5 ± 1.2
	16	98.01 ± 0.46	97.11 ± 0.67	0.90	27.6 ± 1.6	27.2 ± 1.2	24.1 ± 1.4
	64	98.00 ± 0.43	97.10 ± 0.76	0.90	24.6 ± 1.8	23.9 ± 1.1	19.9 ± 1.2
	256	97.91 ± 0.43	97.14 ± 0.72	0.77	10.7 ± 0.9	10.1 ± 0.8	8.7 ± 0.8
DenseNet-121	0	98.24 ± 0.32	97.52 ± 0.67	0.72	29.5 ± 1.3	29.2 ± 1.3	23.7 ± 1.3
	4	98.27 ± 0.30	97.53 ± 0.64	0.73	29.1 ± 1.1	28.9 ± 1.2	22.8 ± 1.2
	16	98.17 ± 0.35	97.41 ± 0.73	0.75	26.2 ± 1.3	26.4 ± 0.9	18.8 ± 1.3
	64	98.19 ± 0.25	97.54 ± 0.78	0.65	17.1 ± 0.8	17.0 ± 1.2	10.8 ± 1.1
	256	97.99 ± 0.35	97.34 ± 0.74	0.65	3.7 ± 0.6	4.4 ± 0.6	2.6 ± 0.6
EfficientNet-B0	0	98.04 ± 0.54	97.30 ± 0.88	0.74	22.3 ± 1.1	22.8 ± 1.3	20.1 ± 1.2
	4	98.05 ± 0.53	97.30 ± 0.87	0.75	22.0 ± 1.0	22.4 ± 1.4	19.5 ± 1.2
	16	98.08 ± 0.55	97.30 ± 0.91	0.78	20.6 ± 1.3	20.8 ± 1.4	17.3 ± 1.2
	64	98.06 ± 0.48	97.26 ± 0.94	0.80	16.1 ± 1.2	15.7 ± 1.3	12.0 ± 1.1
	256	97.99 ± 0.49	97.29 ± 0.75	0.70	5.5 ± 0.8	5.5 ± 0.9	4.2 ± 0.6
MobileNet-V2	0	97.41 ± 0.48	96.61 ± 0.88	0.80	20.6 ± 1.2	21.0 ± 0.9	19.3 ± 1.1
	4	97.41 ± 0.52	96.62 ± 0.91	0.78	20.2 ± 1.0	20.7 ± 1.1	18.3 ± 1.0
	16	97.33 ± 0.58	96.59 ± 0.91	0.75	18.5 ± 1.2	18.7 ± 1.0	15.5 ± 1.2
	64	97.38 ± 0.56	96.63 ± 0.85	0.75	13.7 ± 1.0	13.5 ± 0.9	10.8 ± 0.9
	256	97.19 ± 0.49	96.47 ± 0.80	0.72	4.2 ± 0.8	4.2 ± 0.6	3.5 ± 0.6
ViT-B/16	0	97.83 ± 0.75	96.83 ± 0.80	1.00	28.8 ± 1.2	29.4 ± 1.2	25.1 ± 1.3
	4	97.86 ± 0.71	96.81 ± 0.74	1.05	28.1 ± 1.3	28.7 ± 1.1	23.9 ± 1.4
	16	97.83 ± 0.72	96.78 ± 0.82	1.05	25.3 ± 1.4	25.8 ± 1.1	19.9 ± 1.3
	64	97.88 ± 0.68	96.73 ± 0.76	1.14	17.6 ± 1.3	17.7 ± 1.4	12.7 ± 1.2
	256	97.80 ± 0.72	96.74 ± 0.70	1.06	6.1 ± 0.7	6.3 ± 1.0	4.6 ± 0.6
ConvNeXt-T	0	98.20 ± 0.36	97.30 ± 0.72	0.90	29.3 ± 1.3	29.7 ± 1.5	26.3 ± 1.6
	4	98.17 ± 0.40	97.28 ± 0.67	0.90	28.8 ± 1.3	29.0 ± 1.4	25.1 ± 1.6
	16	98.15 ± 0.44	97.24 ± 0.80	0.91	25.5 ± 1.2	25.6 ± 1.5	20.8 ± 1.5
	64	98.14 ± 0.37	97.27 ± 0.73	0.87	16.8 ± 1.1	16.6 ± 1.1	13.0 ± 0.9
	256	98.10 ± 0.38	97.28 ± 0.74	0.81	5.1 ± 0.6	5.3 ± 0.7	4.0 ± 0.6
Swin-T	0	98.17 ± 0.50	97.20 ± 0.81	0.97	29.0 ± 1.4	28.8 ± 1.5	25.5 ± 1.4
	4	98.12 ± 0.49	97.18 ± 0.80	0.94	28.7 ± 1.4	28.4 ± 1.4	24.9 ± 1.3
	16	98.19 ± 0.48	97.23 ± 0.77	0.96	26.5 ± 1.1	26.2 ± 1.5	22.0 ± 1.4
	64	98.12 ± 0.52	97.34 ± 0.79	0.78	19.6 ± 1.3	19.2 ± 1.3	15.0 ± 1.2
	256	98.14 ± 0.41	97.17 ± 0.61	0.96	6.5 ± 0.8	6.7 ± 0.8	5.2 ± 0.8
RegNetY-1.6GF	0	97.87 ± 0.46	96.91 ± 0.83	0.96	21.0 ± 1.4	21.1 ± 1.1	19.4 ± 1.2
	4	97.80 ± 0.42	96.85 ± 0.83	0.96	20.5 ± 1.3	20.8 ± 1.4	18.9 ± 1.1
	16	97.82 ± 0.49	96.73 ± 0.85	1.09	18.5 ± 1.4	18.4 ± 1.3	15.8 ± 1.1
	64	97.83 ± 0.44	96.72 ± 0.92	1.10	12.3 ± 1.1	11.9 ± 1.0	9.9 ± 1.0
	256	97.72 ± 0.42	96.62 ± 0.85	1.10	3.3 ± 0.7	3.4 ± 0.6	2.7 ± 0.7
EfficientNetV2-S	0	97.70 ± 0.50	96.86 ± 1.01	0.84	19.5 ± 1.4	20.4 ± 1.5	17.6 ± 1.1

Architecture	Rank	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)	NCC
	4	97.69 ± 0.48	96.93 ± 0.93	0.75	19.5 ± 1.7	20.2 ± 1.3	15.6 ± 3.1
	16	97.70 ± 0.51	96.86 ± 0.89	0.83	16.7 ± 2.1	17.8 ± 2.0	11.5 ± 4.2
	64	97.64 ± 0.61	96.85 ± 0.97	0.79	11.0 ± 3.3	12.9 ± 2.7	7.6 ± 3.7
	256	97.65 ± 0.51	96.75 ± 0.91	0.90	3.4 ± 1.3	4.4 ± 1.1	2.9 ± 1.1
MaxViT-T	0	97.88 ± 0.46	97.10 ± 0.83	0.78	21.6 ± 1.2	22.6 ± 1.5	19.1 ± 1.3
	4	97.86 ± 0.46	97.12 ± 0.87	0.73	20.9 ± 1.3	22.0 ± 1.5	18.2 ± 1.4
	16	97.80 ± 0.49	97.07 ± 0.86	0.73	18.3 ± 1.2	19.2 ± 1.7	14.8 ± 1.4
	64	97.90 ± 0.49	97.10 ± 0.86	0.79	12.4 ± 1.0	12.6 ± 1.7	9.4 ± 1.0
	256	97.80 ± 0.42	97.12 ± 0.79	0.68	3.8 ± 0.5	4.0 ± 0.9	3.1 ± 0.6
DINOv2 ViT-S/14	0	95.91 ± 0.75	93.96 ± 1.01	1.96	30.8 ± 0.9	30.3 ± 1.4	26.9 ± 1.3
	4	95.81 ± 0.79	93.93 ± 1.02	1.88	30.5 ± 0.9	30.0 ± 1.4	26.1 ± 1.4
	16	95.70 ± 0.76	93.76 ± 0.97	1.94	28.4 ± 1.1	28.9 ± 1.3	23.8 ± 1.4
	64	95.65 ± 0.73	93.91 ± 1.00	1.74	23.2 ± 1.4	23.0 ± 1.4	16.8 ± 1.5
	256	95.33 ± 0.98	93.12 ± 1.17	2.21	9.3 ± 0.8	9.8 ± 1.2	6.3 ± 1.0

Table 4.8: Amendment 8 probe-matched linear adversary (10 seeds): accuracy (percent), gap (pp), and recoverability R (percent; device probe). λ above the regime collapses P2 accuracy.

Architecture	λ	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)
ResNet-50	0	97.63 ± 0.48	96.77 ± 0.71	0.86	24.7 ± 1.2	25.0 ± 0.9
	0.1	97.12 ± 0.29	96.28 ± 0.44	0.84	9.9 ± 1.1	10.2 ± 0.9
	0.3	96.80 ± 0.78	95.89 ± 0.73	0.91	8.1 ± 1.1	9.0 ± 1.1
	1	94.20 ± 1.88	93.21 ± 1.61	0.99	7.4 ± 1.3	8.1 ± 1.4
	3	83.67 ± 8.43	81.89 ± 8.69	1.78	7.0 ± 1.6	6.7 ± 1.3
DenseNet-121	0	97.68 ± 0.55	96.95 ± 0.35	0.73	29.4 ± 1.3	29.3 ± 1.6
	0.1	97.04 ± 0.60	96.81 ± 0.77	0.22	16.7 ± 1.4	17.2 ± 1.5
	0.3	96.63 ± 1.10	96.05 ± 1.30	0.58	18.3 ± 1.5	17.9 ± 1.0
	1	91.48 ± 2.44	87.96 ± 5.59	3.52	16.6 ± 1.0	16.5 ± 1.8
	3	65.93 ± 7.69	66.11 ± 9.63	-0.19	15.9 ± 2.6	17.0 ± 1.1
ConvNeXt-T	0	97.77 ± 0.73	96.54 ± 0.75	1.22	29.7 ± 1.4	30.0 ± 1.1
	0.1	97.56 ± 0.75	96.56 ± 0.78	1.00	19.0 ± 1.4	19.1 ± 1.6
	0.3	97.58 ± 0.65	96.02 ± 1.80	1.56	17.2 ± 1.2	17.4 ± 1.2
	1	97.00 ± 0.69	95.63 ± 1.18	1.37	16.5 ± 1.9	17.5 ± 1.5
	3	88.36 ± 3.04	71.95 ± 28.54	16.41	14.0 ± 1.7	14.3 ± 3.7

Table 4.9: Amendment 8 Reinhard stain normalisation (10 seeds; strength 0 is the matching Phase 1 run): accuracy (percent), gap (pp), and R (percent).

Architecture	s	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)
ResNet-50	0	97.63 ± 0.48	96.77 ± 0.71	0.86	24.7 ± 1.2	25.0 ± 0.9
	0.5	97.31 ± 0.58	96.12 ± 0.68	1.19	21.6 ± 0.7	21.4 ± 0.9
	1	95.83 ± 0.39	95.01 ± 1.40	0.81	18.0 ± 1.0	18.5 ± 1.1
DenseNet-121	0	97.68 ± 0.55	96.95 ± 0.35	0.73	29.4 ± 1.3	29.3 ± 1.6
	0.5	97.51 ± 0.47	96.30 ± 0.61	1.21	28.2 ± 1.1	28.6 ± 1.4
	1	96.20 ± 0.83	95.30 ± 0.92	0.90	24.5 ± 1.0	25.1 ± 0.9
EfficientNet-B0	0	97.99 ± 0.39	97.07 ± 0.52	0.91	22.1 ± 1.2	22.7 ± 1.1
	0.5	97.37 ± 0.56	96.54 ± 0.60	0.83	21.0 ± 1.6	21.2 ± 1.6
	1	96.21 ± 0.56	95.59 ± 1.13	0.62	18.0 ± 1.2	18.2 ± 0.9
ConvNeXt-T	0	97.26 ± 0.63	96.73 ± 0.46	0.53	28.7 ± 1.2	29.9 ± 1.5
	0.5	97.01 ± 0.76	95.54 ± 1.44	1.47	27.0 ± 2.0	27.9 ± 1.0
	1	96.11 ± 1.24	95.47 ± 1.06	0.64	25.2 ± 1.9	25.4 ± 1.1

4.6 Grad-CAM sees no difference

Leak and clean models placed the same share of Grad-CAM attribution outside the cell mask (21.8% on average, against a background pixel share of 34.6%; leak minus clean -0.09 pp, -0.57 to 0.39). The maps

depended on the trained head (median Spearman 0.08 with a random-head map), so the absence of a difference is not a failure of the method's sanity check.

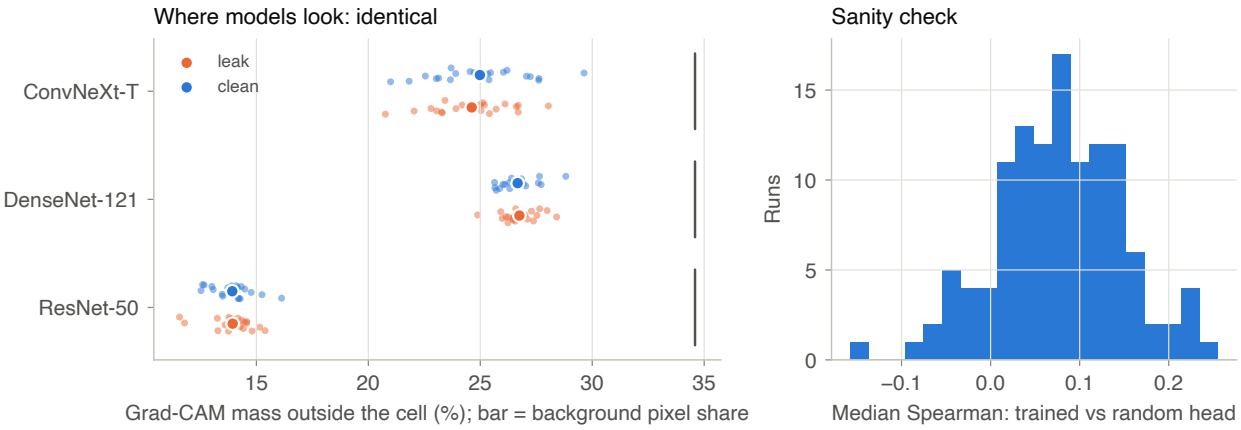


Figure 4.10: Left: per-run Grad-CAM attribution outside the cell for leak and clean models (bar: background pixel share). Right: agreement of each run's maps with random-head maps.

Table 4.10: Grad-CAM (descriptive): share of positive attribution outside the cell mask and background pixel share (percent, 20 runs each), and the median Spearman correlation with a random-head map.

Architecture	Models	Attribution outside	Background pixels	Random-head ρ
ResNet-50	leak	13.9 ± 1.0	34.6 ± 0.5	0.085
	clean	13.9 ± 0.9	34.6 ± 0.5	0.115
DenseNet-121	leak	26.7 ± 0.8	34.6 ± 0.5	0.044
	clean	26.7 ± 0.8	34.6 ± 0.5	0.057
ConvNeXt-T	leak	24.6 ± 1.8	34.6 ± 0.5	0.077
	clean	25.0 ± 2.2	34.6 ± 0.5	0.076

5

The mechanism by elimination

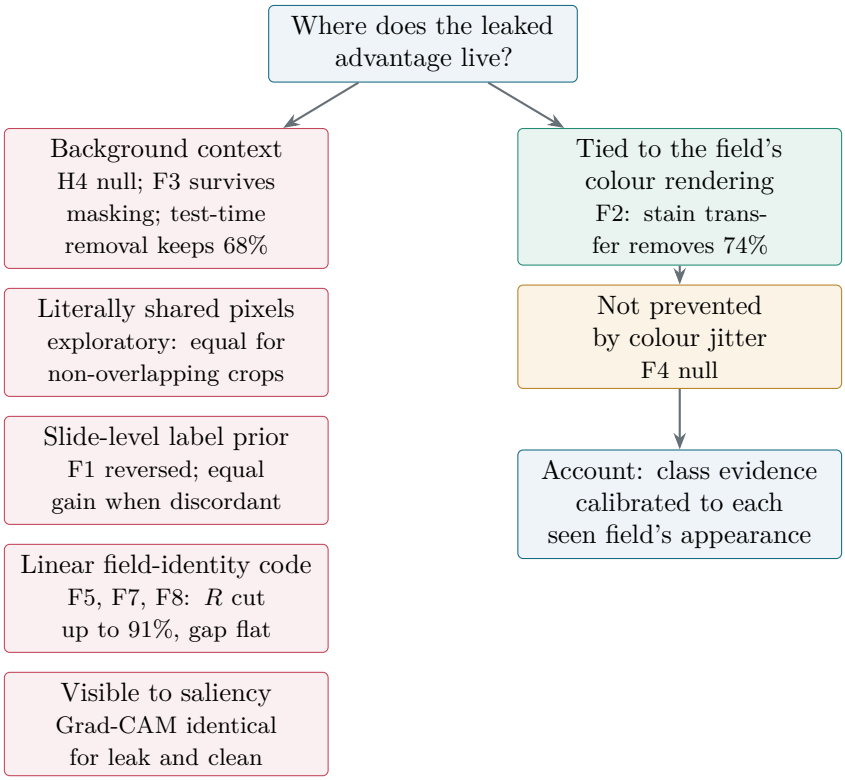


Figure 5.1: The mechanism by elimination. Red: candidate carriers ruled out; green: the one intervention that removed most of the advantage; gold: the qualification that keeps the account modest.

Taken together with Study 2's H4 and H5, the follow-up narrows how models use the leak. The advantage does not need a background (F3, test-time removal, H4). It is not memorisation of shared pixels (exploratory). It is not a slide-level label prior (F1, reversed). It is not carried by a separable, linearly decodable field code: erasing that code in closed form, or suppressing it with an adversary that cannot hide it from a linear probe, cut recoverability by 38–91% without moving the gap (F5, F7, F8). What remains is that the advantage depends on the cell being rendered as its own field renders cells: matching a test cell's colour statistics to another field of the same class removed most of it (F2).

The simplest account consistent with all of this is that a model trained on a field's siblings calibrates its class evidence to that field's appearance (stain, colour balance, and the texture they produce in the cell itself), so cells from a seen field sit further inside their class region. Random colour jitter during training

did not prevent this (F4), presumably because jitter perturbs each image around its own appearance and leaves the field-specific structure of that appearance learnable. This account is a reading of the evidence, not a registered result, and the F2–F4 tension is the part of it most in need of a direct test (for example, stain transfer applied to colour-augmented models).

Decodable is not used, and saliency is not use

Study 1 established *availability*: a probe recognised fields at 109 times chance. The follow-up shows that availability and use come apart: the identity a probe reads can be removed almost entirely without touching the inflation, in line with evidence from language models that probe-decodable information need not be what a model relies on [3, 4]. Grad-CAM showed leak and clean models attending to the same places; a saliency audit would have cleared both.

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